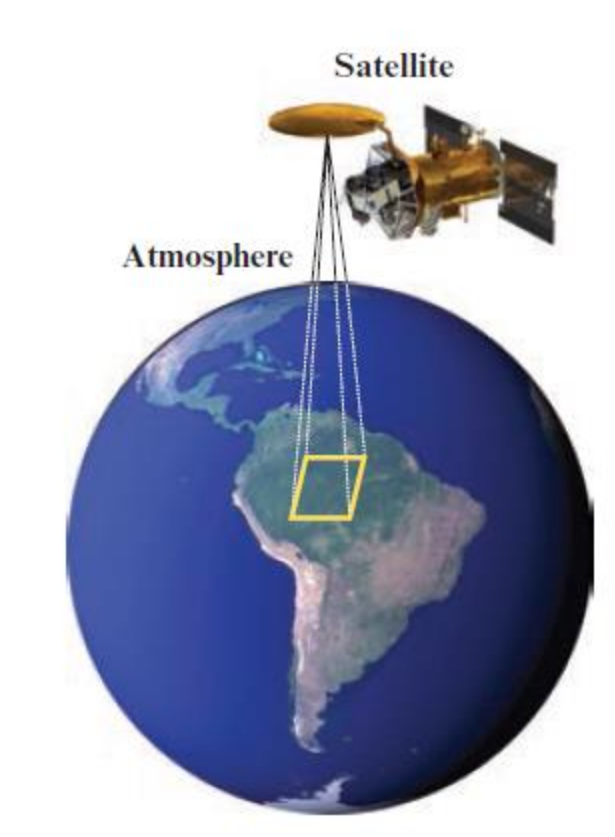

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REMOTE SENSING SENSORS AND DATA ACQUISITION



MULTISPECTRAL/ Hyperspectral RS

- **Multispectral Remote Sensing:**
  - **Collection of reflected, emitted, or back-scattered energy from an object or area of interest in multiple bands (regions) of the electromagnetic spectrum.**
- **Hyperspectral Remote Sensing:**
  - **Involves data collection in hundreds of bands.**

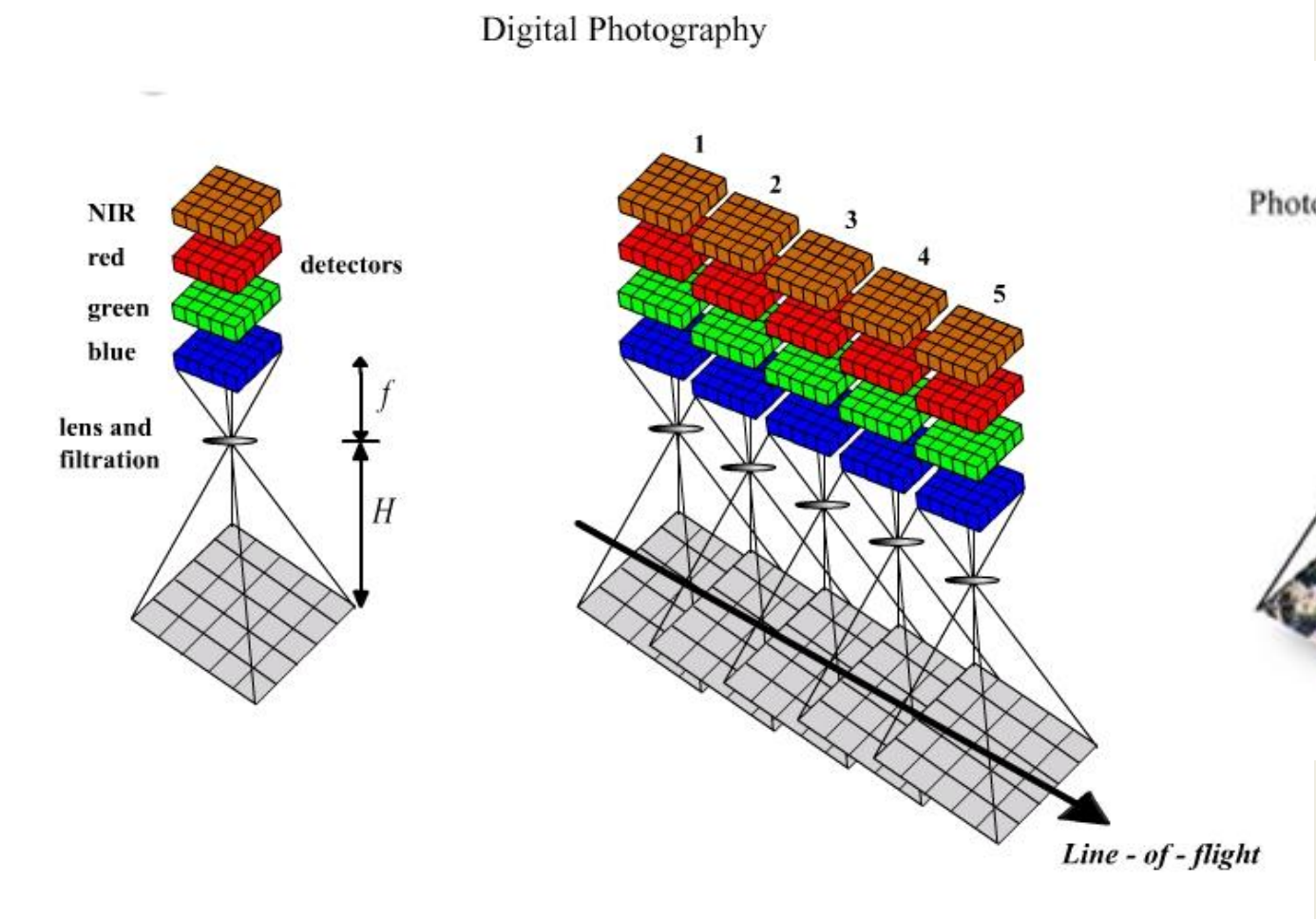
REMOTE SENSORS

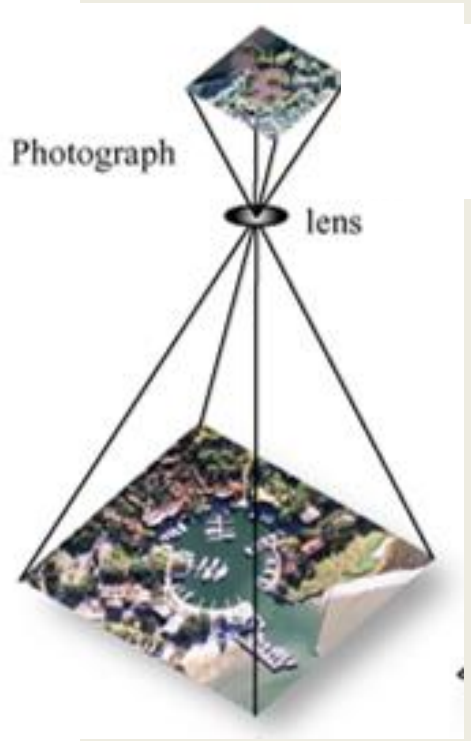
- ***Remote Sensors:*** **The detection and recording instruments for Remote Sensing**
  - **Photographic cameras (Analog and Digital)**
  - **Electro-Mechanical Scanners**
- ***Two main types***
  - **Framing Systems**
  - **Scanning Systems**

MULTISPECTRAL SCANNER/ SENSORS

- Scanning system that use to collect data over different wavelength ranges is called a multispectral scanner (MSS)
- Common Remote Sensing acquisition technologies
  - Whisk Broom (Across Track- Oscillating Mirror)
  - Push Broom (Along Track- Linear array of detectors)

Frame Grabber Technology





http://www.eomonline.com/EOM_Jul05/article.php?Article=department3

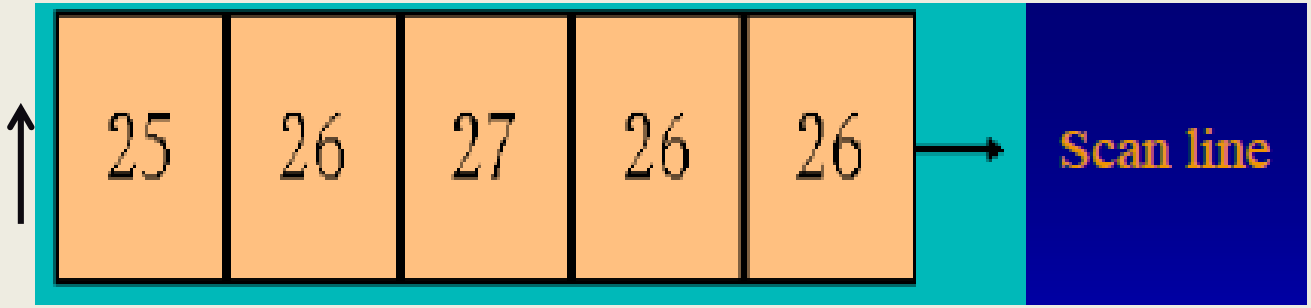
Aerial Photos



Frame Camera Image

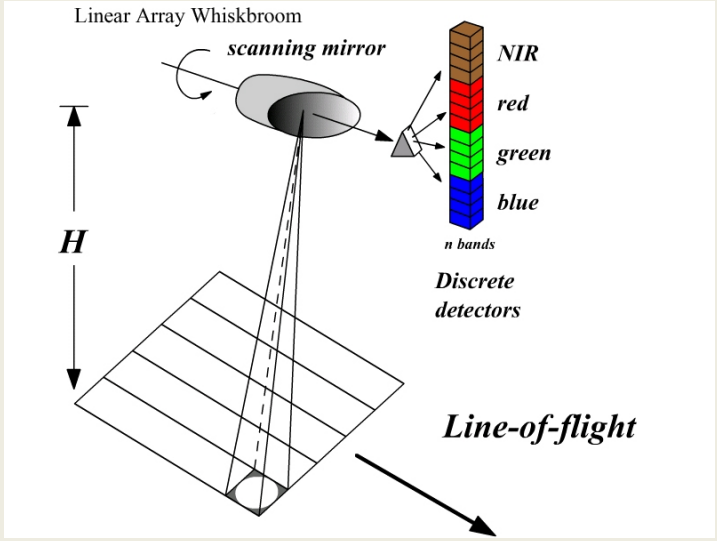
Scan Line

- A horizontal row of pixels in an image
- Scan line is collected either
  - Sequentially as the sensor moves left to right (whiskbroom) or
  - Simultaneously using a linear array of photo-detectors (frame grabber, push broom)



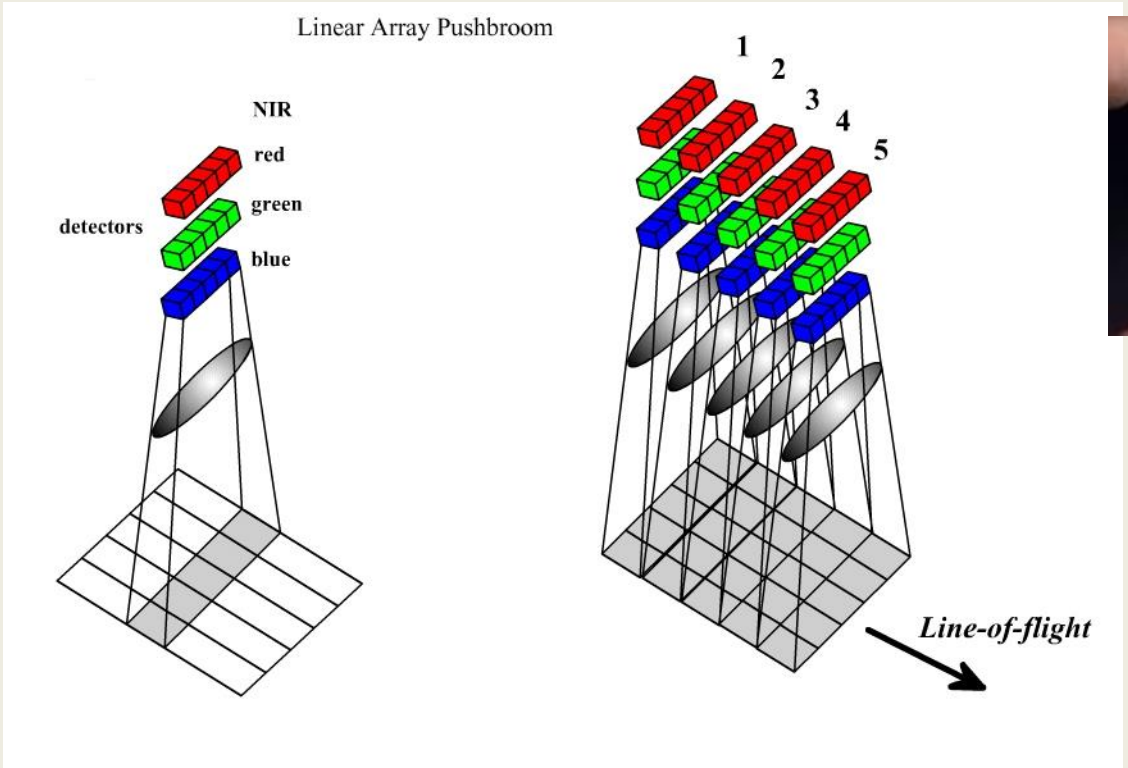
Whisk broom- Across track

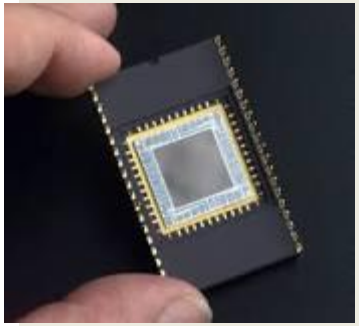
- Utilizes a mirror that sweeps in a direction perpendicular to the flight path
- EMR is directed via the optics into a series of mirrors and prisms
- EMR split into its various wavelengths and focused onto detectors
- Detectors measure EMR from a point on the ground and stores



Push Broom- Along Track

- Uses charged couple devices aligned in a direction perpendicular to the flight direction
- Each CCD aimed at a specific point on the ground with the neighboring CCD viewing the neighboring ground location
- Entire line of image data (line of pixels) acquired simultaneously.

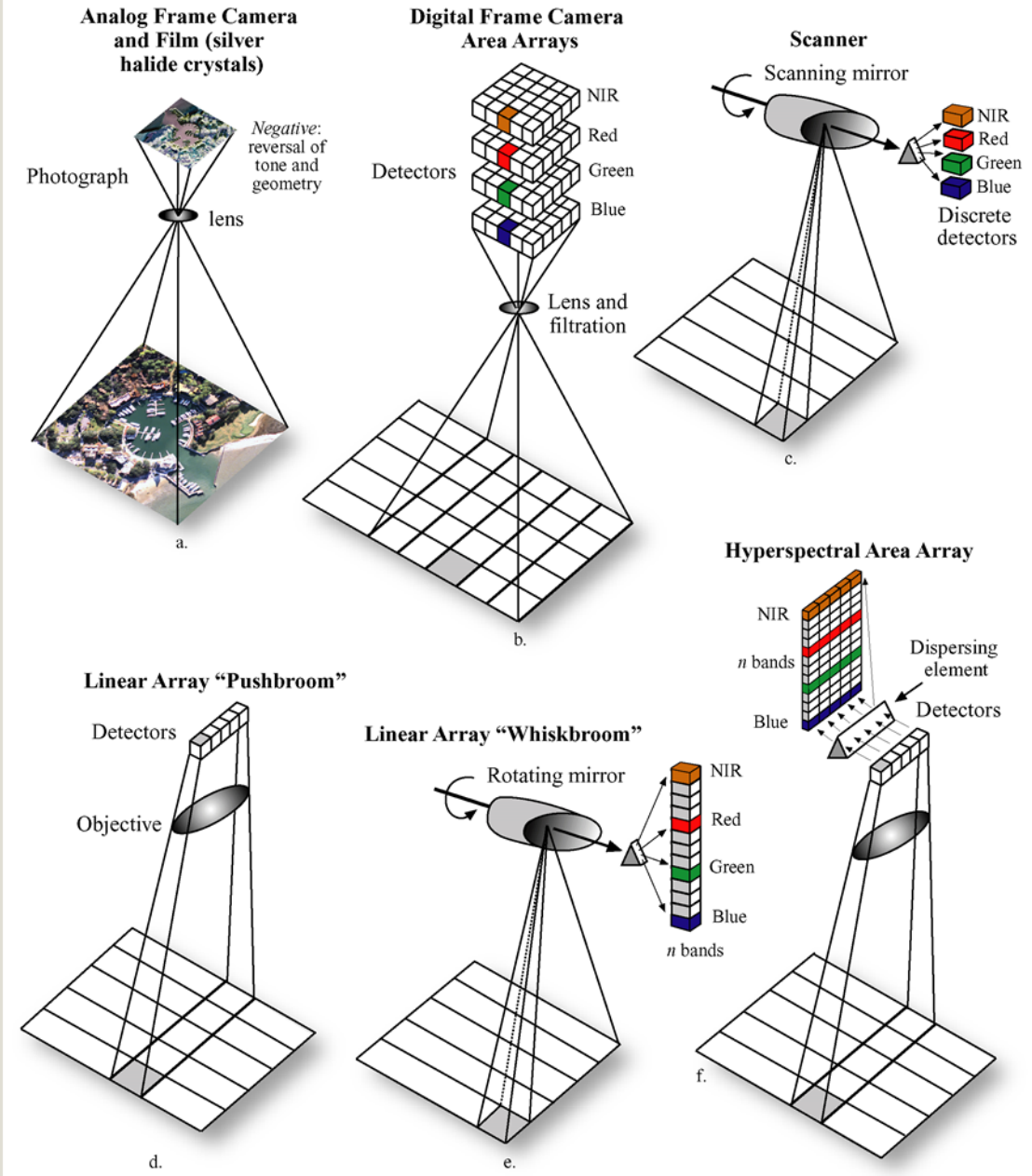


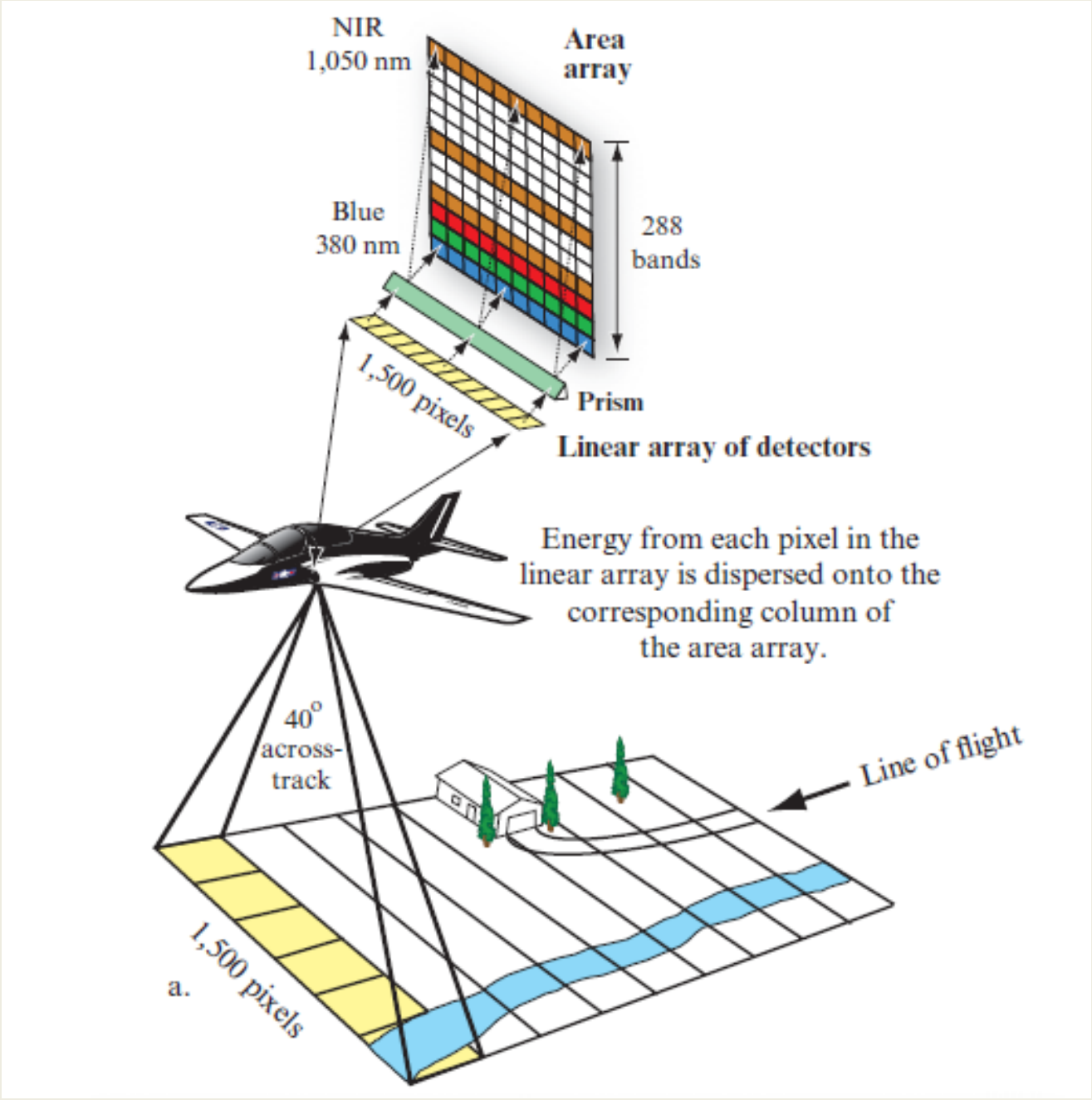


Imaging Geometry

- **Point Geometry**
  - **Whisk Broom (Across Track- Oscillating Mirror (Point sensors)**
  - **Images only a single point at any one instant of time**
- **Line Geometry**
  - **Push Broom (Along Track- Linear array of detectors, (Line sensors)**
  - **Images a complete scan line at any one instant of time**

Remote Sensing Systems Used to Collect Aerial Photography, Multispectral and Hyperspectral Imagery




NIR
1,050 nm
Area
array
Blue
380 nm
288
bands
1,500 pixels
Prism
Linear array of detectors
Energy from each pixel in the linear array is dispersed onto the corresponding column of the area array.
40°
across-
track
Line of flight
1,500 pixels
a.

Airborne Digital Cameras

Leica Geosystems

- **Airborne Digital Sensor System ADS-40/80**
- **4 bands (RGB NIR**
- Orientation of the aft, Nadir, and forward- looking linear arrays



RedEdge Micasense

- 8 cm / pixel
- 5 narrow bands
- 12 Bit sensor
- Light weight (168 grams)
- Built in Wi/Fi





Parrot Sequoia

- 13cm / pixel
- 4 narrow bands (G, R , Red Edge, NIR)
- 12 Bit sensor
- Light weight (107 grams)
- Built in Wi/Fi



Altum- Micasense

A revolutionary 3-in-1 solution…

One sensor

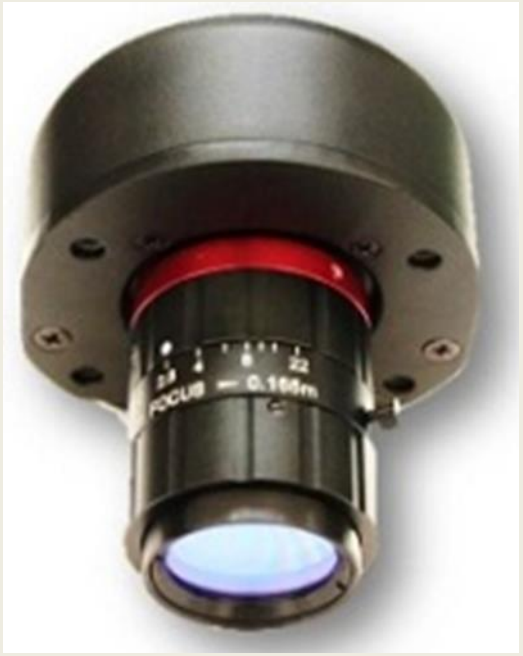
Thermal, multispectral, and high-res RGB imagery

- **4cm / pixel**
- **5 narrow bands** (Blue, green, red, red edge, near infrared (NIR),
- **Thermal infrared 8-14 um (81 cm)**
- **16 Bit sensor**
- **Weight (406 grams)**



OCI UAV Airborne Hyperspectral Camera

- 272 grams
- 120 bands
- Wavelength Range 460-970 nm
- Push-broom



Dwell Time

- **The time the Instantaneous Field of View (IFOV) "sees" a ground**
- **Generally, quite short and influences the design of the spatial, spectral, and radiometric resolutions of the sensor**

Spatial, Spectral, and Radiometric Resolutions

- There are trade-offs between spatial, spectral, and radiometric resolutions
- For high spatial resolution, the sensor has to have a small IFOV
- This reduces the amount of energy that can be detected as the area of the ground resolution cell within the IFOV becomes small
- Leads to reduced radiometric resolution-the ability to detect fine energy differences

Spatial, Spectral, and Radiometric Resolutions

- To increase the amount of energy detected (and thus, the radiometric resolution) without reducing spatial resolution, would have to broaden the wavelength range (Reduce Spectral Resolution)
- Conversely, coarser spatial resolution would allow improved radiometric and/or spectral resolution
- Thus, these three types of resolution must be balanced against the desired capabilities and objectives of the sensor

Data Transmission



Data Transmission

- For Aircraft, data is retrieved as it lands, processed and delivered
- For Satellite, data need to be electronically transmitted to ground station, and is done as
  - Direct Transmission if Satellite and Ground Stations are in the line of sight
  - Data can be recorded on board and transmitted at later time

Data Transmission

- Data can also be relayed to the Ground Receiving Station through the Tracking and Data Relay Satellite System (TDRSS)
  - TDRSS: a series of communications satellites in geosynchronous orbit
  - The data are transmitted from one satellite to another until they reach the appropriate Ground Receiving Station

